

# Acute myeloid leukemia

≥20% myeloid blasts — or any blast % with a defining genetic lesion · classification and treatment are now genetically driven · every new diagnosis is triaged for the three emergencies before a regimen is chosen

Compiled from the Encyclopedia AML review note and Fellowship Anki deck · ELN 2022 · WHO 2022 / ICC · NCCN. Items marked † extend beyond the existing review note.

## DIAGNOSIS

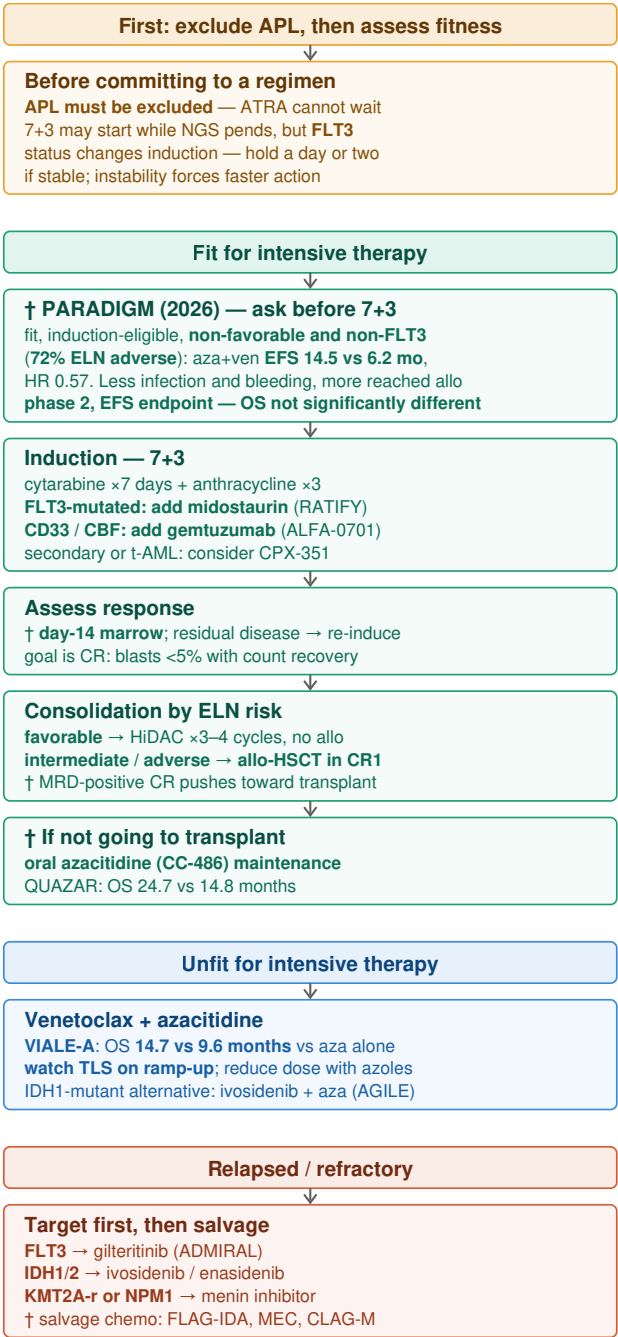
- ≥20% myeloid blasts in blood or marrow — or any blast % with a defining genetic lesion: t(8;21), inv(16)/t(16;16), PML::RARA, and (WHO 2022) NPM1 or CEBPA-bZIP
- Also AML at <20%: **myeloid sarcoma** / extramedullary disease
- “Blasts” on smear is morphologic — **confirm lineage by flow**
- Workup: marrow aspirate + biopsy with **flow, karyotype + FISH, NGS**; CBC/smear (Auer rods); **coags + fibrinogen + D-dimer**; TLS labs; **baseline echo**; HLA typing early; LP only if CNS symptoms or monocytic, after coags corrected
- Stat **FISH/PCR for PML::RARA** on any suspicion of APL
- † Median age ~68. The three triage questions on every new AML: **Is it APL? Is there leukostasis? Is there tumor lysis?**

## ELN 2022 GENETIC RISK

| Risk         | Representative genetics                                                                                                                                                                                  |
|--------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Favorable    | t(8;21) RUNX1::RUNX1T1 · inv(16)/t(16;16) CBFB::MYH11 ( <b>core-binding factor</b> ) · <b>mutated NPM1 without FLT3-ITD</b> · in-frame bZIP CEBPA                                                        |
| Intermediate | NPM1 + FLT3-ITD · wild-type NPM1 + FLT3-ITD · t(9;11) MLLT3::KMT2A · anything not otherwise classified                                                                                                   |
| Adverse      | Complex or monosomal karyotype · -5/del(5q), -7, abn(17p) · <b>TP53</b> · MDS-related genes (ASXL1, BCOR, EZH2, RUNX1, SF3B1, SRSF2, STAG2, U2AF1, ZRSR2) · KMT2A rearrangements · <b>t(6;9), inv(3)</b> |

ELN 2022 changes worth knowing: FLT3-ITD is intermediate regardless of allelic ratio (ratio dropped) · only **in-frame bZIP CEBPA** is favorable · **NPM1** with adverse cytogenetics is reclassified adverse · **KIT** co-mutation worsens core-binding-factor AML.

## TREATMENT PATHWAY



## REGIMENS & AGENTS

| Regimen                  | Notes                                                                                                                                                                                                                                                                                                            |
|--------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7+3                      | <b>Cytarabine 100–200 mg/m<sup>2</sup>/day</b> continuous infusion ×7 days († <b>SHC uses 100</b> ) + <b>daunorubicin 60 mg/m<sup>2</sup> IV days 1–3</b> . E1900 showed <b>90 &gt; 45</b> , but † <b>AML17 found 90 vs 60 gave no CR or OS benefit and doubled 60-day mortality</b> — 60 is the modern standard |
| + midostaurin            | FLT3-mutated (RATIFY). Quizartinib for FLT3-ITD (QuANTUM-First)                                                                                                                                                                                                                                                  |
| + gemtuzumab             | CD33-positive, fractionated dosing; greatest benefit in <b>core-binding factor</b> (ALFA-0701)                                                                                                                                                                                                                   |
| CPX-351                  | Liposomal cytarabine + daunorubicin for t-AML / AML-MRC. <b>Caveat</b> : comparator was 7+3, not ven/aza — niche today is the fitter older adult bridging to allo                                                                                                                                                |
| HiDAC                    | 3 g/m <sup>2</sup> q12h ×3–4 cycles; best in CBF. <b>Age &gt;60 → IDAC 1.5 g/m<sup>2</sup></b> — 3 g/m <sup>2</sup> is inappropriate (ocular, cerebellar, infectious toxicity). Renal dose-reduce                                                                                                                |
| Venetoclax + azacitidine | Unfit patients (VIALE-A). Ramp-up with TLS prophylaxis; <b>reduce venetoclax with CYP3A inhibitors</b>                                                                                                                                                                                                           |
| † Salvage                | FLAG-IDA · MEC · CLAG-M · clinical trial · second transplant                                                                                                                                                                                                                                                     |

## † MRD AND THE TRANSPLANT DECISION

- **Measure MRD at the end of every cycle** and again before transplant — order **HemeSTAMP MRD** (NGS) + **multiparameter flow cytometry**; add **NPM1 PCR** where applicable
- **MRD-positive CR is not CR for planning purposes** — it predicts relapse and moves an otherwise favorable-risk patient toward transplant
- **Allo-HSCT in CR1**: generally **not** for favorable risk · **yes** for intermediate and adverse risk · modified by MRD, age, comorbidity, and donor availability
- Allo cures through **graft-versus-leukemia**; **autologous transplant is not used** — the graft carries the same mutations
- Fitness for intensive therapy is a composite of age, performance status and comorbidity (HCT-CI) — not age alone

Acute myeloid leukemia

THE THREE EMERGENCIES

APL — drop everything

↓

Recognize

DIC + bleeding + hypergranular promyelocytes  
faggot cells (bundled Auer rods)  
microgranular variant can look monocytic

↓

Start ATRA immediately

do **not** wait for PML::RARA confirmation  
early death from hemorrhage is the leading cause of APL mortality — and preventable

↓

Support aggressively

fibrinogen >100–150 (cryoprecipitate)  
platelets >30–50k; INR near normal  
coags q6–12 h; avoid LP and central lines

↓

Then treat

ATRA + arsenic, chemo-free, low/int risk  
high risk (WBC >10k): add anthracycline  
cure >90%; **do NOT leukapheresis — worsens DIC**

Leukostasis

↓

Hyperleukocytosis + symptoms

myeloid blasts are large and sticky — symptoms at lower counts than ALL/CLL, often >50–100k  
CNS (headache, confusion, vision) + **pulmonary** (hypoxia)

↓

† Management — with doses

hydroxyurea 50–100 mg/kg/day PO, divided q6–12h (≈2–3 g q12h) until WBC <25k; mucositis ≥75 mg/kg/day HU takes days — if too slow, **cytarabine 100 mg/m²/day** urgent induction is definitive; apheresis if severe — **never in APL**, no mortality benefit. **No RBC transfusion** hydrate + TLS ppx. † **dex 10 mg IV q6h** if monocytic + ARF

Tumor lysis

↓

Cairo-Bishop

lab TLS = ≥2 of ↑K, ↑phosphate, ↑uric acid, ↓Ca  
clinical TLS = lab TLS + AKI, arrhythmia or seizure

↓

Prevent and treat

aggressive IV hydration is the cornerstone  
allopurinol 300 mg BID standard risk  
rasburicase if high risk or uric acid >8 — **not in G6PD**  
no urinary alkalinization

DRUG TOXICITY & MONITORING

| Agent                     | Watch for                                                                                                                                                                                                     |
|---------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Cytarabine                | Pyrimidine analog. High-dose: <b>cerebellar toxicity — cerebellar exam before every HiDAC dose, hold if abnormal; conjunctivitis → prophylactic steroid eye drops</b> ; ara-C syndrome (fever, myalgia, rash) |
| Anthracyclines            | Cumulative, dose-dependent <b>cardiotoxicity</b> — track total dose, baseline ± serial LVEF. <b>Vesicant</b> — central access                                                                                 |
| Venetoclax                | <b>Tumor lysis on ramp-up</b> ; neutropenia; <b>dose-reduce with azole antifungals</b> (CYP3A)                                                                                                                |
| Azacitidine vs decitabine | Both hypomethylating; <b>azacitidine incorporates into RNA and DNA</b> , decitabine DNA-only. Azacitidine: injection-site reactions                                                                           |
| Gemtuzumab                | Anti-CD33 calicheamicin ADC — <b>hepatotoxicity and VOD/SOS</b> , especially peri-transplant; fractionated dosing lowers risk                                                                                 |
| Arsenic trioxide          | <b>QT prolongation</b> — monitor ECG, replete K and Mg                                                                                                                                                        |
| Differentiation syndrome  | ATRA/ATO <b>and also IDH and menin inhibitors</b> . Fever, dyspnea, hypoxia, infiltrates, effusions, weight gain, AKI → <b>dexamethasone 10 mg IV BID at first suspicion</b>                                  |
| Menin inhibitors          | Revumenib, ziftomenib — differentiation syndrome + <b>QT prolongation</b>                                                                                                                                     |

SUPPORTIVE CARE

- **Transfusion**: platelets >10k (>20–50k if bleeding, procedure, or APL); **irradiated and leukoreduced** products
- † **Prophylaxis during induction**: antibacterial (levofloxacin), **antifungal (posaconazole)**, antiviral (acyclovir), PJP coverage — remember posaconazole forces a venetoclax dose reduction
- **Febrile neutropenia**: fever ≥38.3°C once (or ≥38.0°C for 1 h) with ANC <500 → **antipseudomonal beta-lactam within 1 hour**. Add vancomycin for instability, line infection, skin/soft tissue, severe mucositis or pneumonia. Antifungal if fever persists >4–7 days
- Fertility counseling before chemotherapy; central venous access; early palliative care in older or adverse-risk disease

TRIAL EVIDENCE

|                         |                                                                                                                                                                                                                               |
|-------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>RATIFY</b>           | Midostaurin + 7+3 in FLT3-mutated — survival benefit                                                                                                                                                                          |
| <b>QuANTUM-First</b>    | Quizartinib + 7+3 in FLT3-ITD                                                                                                                                                                                                 |
| <b>ALFA-0701</b>        | Fractionated gemtuzumab + 7+3 in CD33-positive disease                                                                                                                                                                        |
| <b>E1900</b>            | Daunorubicin <b>90 vs 45 mg/m²</b> — better CR and OS in age <60                                                                                                                                                              |
| <b>† AML17</b>          | Daunorubicin <b>90 vs 60 mg/m² — no CR or OS difference</b> , and 60-day mortality <b>10% vs 5%</b> (HR 1.98). <b>Read with E1900: 60 is standard, not 90</b> (Burnett, Blood 2015)                                           |
| <b>VIALE-A</b>          | Azacitidine + venetoclax vs azacitidine — OS <b>14.7 vs 9.6 months</b> (DiNardo, NEJM 2020)                                                                                                                                   |
| <b>† PARADIGM</b>       | <b>Aza+ven vs intensive induction in fit patients</b> (CBF/FLT3/ NPM1 <60 excluded, 72% ELN adverse): EFS <b>14.5 vs 6.2 mo</b> , HR 0.57; less infection and bleeding. <b>Phase 2; OS not significant</b> (Fathi, NEJM 2026) |
| <b>AGILE</b>            | Ivosidenib + azacitidine in IDH1-mutant unfit patients                                                                                                                                                                        |
| <b>ADMIRAL</b>          | Gilteritinib in relapsed FLT3-mutated AML — survival benefit                                                                                                                                                                  |
| <b>APL0406</b>          | ATRA + arsenic beat ATRA + chemo in low/intermediate-risk APL (Lo-Coco, NEJM 2013)                                                                                                                                            |
| <b>† QUAZAR AML-001</b> | Oral azacitidine (CC-486) maintenance in CR1 — OS <b>24.7 vs 14.8 months</b>                                                                                                                                                  |
| <b>AML15</b>            | IDAC vs HiDAC consolidation — no overall survival difference                                                                                                                                                                  |
| <b>SWOG S1203</b>       | IA regimen showed no benefit and more toxicity — <b>7+3 remains standard</b>                                                                                                                                                  |

GENETICS — FINE POINTS

|                        |                                                                                                                                                                          |
|------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>inv(3) / t(3;3)</b> | MECOM (EVI1, 3q26) activation; classically <b>normal or high platelets</b> ; adverse and <b>venetoclax-resistant</b>                                                     |
| <b>t(6;9)</b>          | DEK::NUP214 — associates with <b>FLT3-ITD and basophilia</b> ; adverse                                                                                                   |
| <b>NPM1 flow mimic</b> | NPM1-mutated AML can be <b>HLA-DR– / CD34–</b> and mimic APL — confirm PML::RARA                                                                                         |
| <b>Therapy-related</b> | After <b>alkylators/RT</b> : del(5q)/del(7q) ± TP53, latency ~5–7 yr. After <b>topoisomerase-II</b> : <b>11q23 KMT2A</b> , latency ~2–3 yr                               |
| <b>† Germline</b>      | <b>DDX41, RUNX1, CEBPA, GATA2</b> , telomere biology — send germline testing if young, family history, or multiple cytopenias; <b>it changes related-donor selection</b> |

# Acute myeloid leukemia

## † FOUR WAYS AN ACUTE LEUKEMIC DIES IN THE ED

- **1. Leukostasis** — CVA, MI, respiratory failure
- **2. Bleeding / DIC / thrombosis**
- **3. Infection** — **very low threshold for broad-spectrum antibiotics**
- **4. Tumor lysis** — renal failure, electrolyte derangement

## OVERNIGHT DECISIONS

- **Tests to order** (see next column)
- **Triage floor vs ICU** — **involve the Med 8 fellow:** leukapheresis and CRRT both require ICU
- **Cytoreduction / ATRA**
- **TLS:** start fluids + allopurinol, set lab frequency (**q4h to daily**). **Rasburicase if uric acid >8**
- **DIC:** set lab frequency; cryoprecipitate / FFP as needed
- **Infection:** very low threshold for broad-spectrum antibiotics
- **APL** — discuss with the on-call pathology resident overnight (p 12005)
- **Discuss with the Med 8 attending overnight** for any concern
- **Transfusion medicine / apheresis on call: p 12027**

## WHAT TO SEND

| Test               | When                                                                                                                                                                                                                                         |
|--------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Peripheral studies | <b>Flow cytometry, AML prognosis assay</b> (NPM1, FLT3-ITD, FLT3-TKD), <b>IDH mutation analysis</b> . Send if <b>WBC &gt;10k</b> and marrow will be delayed, or the patient is old enough that you would rather not subject them to a marrow |
| Cytogenetics       | <b>FISH for PML::RARA and BCR::ABL1</b> . Send from <b>peripheral blood</b> if there is any concern for APL or a delay in getting marrow                                                                                                     |
| HLA typing         | Send if the patient is a possible allogeneic transplant candidate                                                                                                                                                                            |

## CYTOREDUCTION OVERNIGHT

| Dose and trigger |                                                                                                                                                                                                                                                                                                                                                                                                          |
|------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Hydroxyurea      | Consider if <b>WBC &gt;20k</b> . Overnight you can order a <b>one-time dose</b> . Typical range <b>500 mg–2 g, max 2 g q6h</b> . † Literature range 50–100 mg/kg/day; <b>grade 3–4 mucositis rises at ≥75 mg/kg/day</b> , so 2 g q6h is the ceiling, not the starting dose                                                                                                                               |
| Leukapheresis    | Consider for <b>evidence of leukostasis</b> — headache, blurry vision, dyspnea/hypoxia, chest pain — typically with <b>blasts &gt;50k in AML</b> (rare in ALL). <b>Page apheresis 12027</b> . Ask ER or the MICU triage fellow for a <b>trialysis line</b> ; admit to MICU. Ask the apheresis RN to <b>save the product in the FGR fridge</b> and coordinate with the research team. <b>Never in APL</b> |
| Cytarabine       | <b>1 g ×1</b> may be considered for leukostasis when apheresis will be delayed                                                                                                                                                                                                                                                                                                                           |

## APL OVERNIGHT

- **Start ATRA on any concern for APL** — do not wait for PML::RARA
- **Smear:** bilobed / folded nuclei ± Auer rods. **Page pathology on call (12005)** if unsure
- **ATRA 45 mg/m²/day PO divided q12h** (= 22.5 mg/m² BID) — overnight you can give a one-time dose
- **DIC labs q6–8h**. Transfuse **platelets >30, fibrinogen >100, INR <2**

## INDUCTION REGIMENS — DOSING

| Regimen           | Dosing                                                                                                                                                                                                                                          |
|-------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7+3               | Cytarabine <b>100–200 mg/m²/day</b> IV continuous infusion ×7 days from day 1 († <b>SHC uses 100</b> ) + daunorubicin <b>60 mg/m²</b> IV daily days 1–3 <i>AML17 / Burnett 2015</i>                                                             |
| 7+3 + GO          | Add gemtuzumab ozogamicin <b>3 mg/m²</b> (max 4.5 mg) IV over 2 h on <b>days 1, 4, 7</b> <i>ALFA-0701 / Lambert 2019</i>                                                                                                                        |
| 7+3 + midostaurin | Add midostaurin <b>50 mg PO q12h on days 8–21</b> <i>RATIFY / Stone 2017</i>                                                                                                                                                                    |
| CPX-351           | Liposomal daunorubicin <b>44 mg/m²</b> + cytarabine <b>100 mg/m²</b> IV on <b>days 1, 3, 5</b> <i>Lancet 2018</i>                                                                                                                               |
| FLAG-IDA-VEN      | Fludarabine <b>30 mg/m²</b> + cytarabine <b>1.5–2 g/m²</b> IV days 2–6; idarubicin IV ( <b>ND-AML 8 mg/m² days 4–6; R/R 6 mg/m² days 4–5</b> ); filgrastim 5 mcg/kg days 1–7; venetoclax ramp-up (100/200/400 mg) days 1–14 <i>DiNardo 2021</i> |
| CLAD / LDAC / VEN | Cladribine <b>5 mg/m²</b> IV days 1–5 + cytarabine <b>20 mg/m²</b> (SQ BID days 1–10; † <b>20 mg/m² IV daily preferred at SHC</b> ) + venetoclax PO days 1–21, alternating with azacitidine <i>Kadia 2022</i>                                   |
| HMA / VEN         | Venetoclax (ramp-up) daily ×28 days + <b>decitabine 20 mg/m² days 1–5</b> or <b>azacitidine 75 mg/m² days 1–7</b> <i>VIALE-A / DiNardo 2020</i>                                                                                                 |

## APL INDUCTION — DOSING

| Regimen          | Dosing                                                                                                                                                                                                                             |
|------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ATO / ATRA       | Arsenic trioxide <b>0.15 mg/kg</b> IV over 2 h daily from day 1 + ATRA <b>22.5 mg/m² PO BID</b> from day 1 — both until hematologic CR or <b>max 60 days</b> <i>APL0406 / Lo-Coco 2013</i>                                         |
| ATO / ATRA / IDA | ATO <b>0.15 mg/kg</b> IV days 9–36 + ATRA <b>22.5 mg/m² BID</b> days 1–36 + idarubicin on days 2, 4, 6, 8 by age: <b>≤61 yr 12 mg/m² · 61–70 yr 9 mg/m² · &gt;70 yr 6 mg/m²</b> <i>APML4 / Iland 2012</i>                          |
| ATO / ATRA / GO  | ATO <b>0.3 mg/kg</b> IV daily days 1–5, then <b>0.25 mg/kg twice weekly weeks 2–8</b> + ATRA <b>22.5 mg/m² BID</b> from day 1 until CR or max 60 days + gemtuzumab <b>6 mg/m²</b> IV once on day 1 <i>AML17 APL / Burnett 2015</i> |